Pipeline Architecture & Technical Post-Mortem

SYSTEM SPECIFICATION: Native 3K HEVC (2880x1620 @ 30 FPS) | Bitrate: ~4.2 Mbps
EXTRACTION ENGINE: Lossless Stream Slicing (>250x Realtime Speed) | Enforced Buffer: +/-60s

1. Technical Root Cause Analysis

Bug 1: Stage 1 Premature Motion Window Cutoff
In Clip 14 (094423.mp4, duration 44m 57s), automated motion detection stopped after 42 seconds (09:44:23 to 09:45:05) because initial entry movement ceased as the operator stepped behind the door frame. Consequently, the re-entry at 09:53:20 and valve operation at 09:53:38 had zero candidate frames extracted. *Resolution:* Implemented continuous rolling temporal sliding windows and multi-tier manual verification.

Bug 2: Database Sorting Inversion Bug
Legacy scripts queried ORDER BY device_id, sequence_number ASC. Because sequence_number was the relative intra-clip second offset (0-2697) rather than absolute wall-clock timestamp, multi-clip queries interleaved detections from different hours, creating inverted ranges (e.g. 11:02 to 09:43) in 29/30 CSV files. *Resolution:* Enforced ORDER BY timestamp ASC, bringing timestamp inversions down to 0 across all 30 days.

2. Calibrated Native 3K Spatial Geometry

Facility Zone ID	Native 3K Bounding Box [x1, y1, x2, y2]	Hazard Category	Operational Purpose
Stage 1 Perimeter	[295, 20, 1489, 780] (1194x760)	**Observation Mask**	Filters sky/terrace noise; covers 100% plant assets
Zone 1A (Skid Valve)	[1080, 370, 1210, 500]	**CRITICAL HAZARD**	Manual backwash lever (Day 1 09:53:38 incident point)
Zone 1B (Filter Heads)	[810, 440, 990, 580]	**CRITICAL HAZARD**	Dual vessel multiport high-pressure heads
Zone 2 (Power Switch)	[995, 340, 1095, 495]	**SEVERE HAZARD**	Main electrical breaker box (Day 1 22:27:36 incident point)
Zone 3 (Skid Work Deck)	[1080, 320, 1420, 790]	**ELEVATED MONITORING**	Technician walkway corridor and control station

3. High-Speed Bitstream Extraction Architecture

By enforcing direct HEVC bitstream copying (-c copy), re-encoding overhead was completely eliminated. The entire 30-day incident dataset of **179 pristine video clips (8.20 GB) was extracted in 108.7 seconds** directly from read-only external storage (>250x realtime speed), with enforced +/-60s pre/post roll safety margins.